Assignment 1 AI

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A-I) Rationality in intelligent agents:

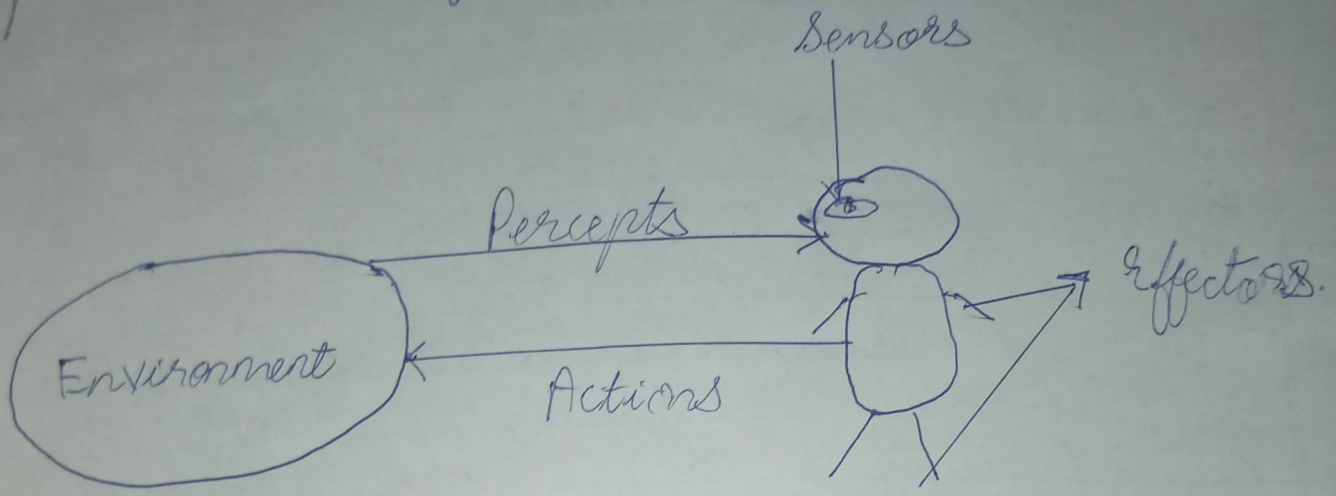
- Rationality epitomizes the cornerstone of intelligent decision-making, where agents meticulously analyze available information to discern the most advantageous course of action.

- It encapsulates a sophisticated process of deliberation, weighing potential outcomes against each other to maximize expected utility or achieve specified objectives.

- Rational agents possess the acumen to navigate through uncertainity, resource constraints, and varying environmental dynamics while maintaining a steadfast focus on their goals.

- Example: Imagine a recommendation system employed by an online retailer. Through rationality, it scrutinizes user preferences, purchase history and product characteristics to offer personalized suggestions that optimize user satisfaction and increase sales revenue.

- Rationality serves as the bedrock upon ~~tow~~ which intelligent agents operate, guidance their actions and strategies to navigate the intricacies of their environments with precision and efficacy.



A-2) Nature of environments for intelligent agents

- Environments encompass the contexts in which intelligent agents operate, comprising various factors that shape ~~of~~ agent behaviour and performance.

- Key characteristics include complexity, dynamism, observability and stochasticity, each posing distinct challenges and opportunities for agents.

- These characteristics profoundly influence the design and functioning of agents, dictating their strategies, decision-making process, and interaction with the surroundings.

- Example: In a medical diagnosis system, the environment is dynamic and stochastic, with patient symptoms changing over time and diagnostic outcomes subject to uncertainty.

- Example: A robot navigating through a cluttered warehouse faces a complex and partially observable environment, necessitating sophisticated sensing, planning, and navigation capabilities.

- Environment types →
  - Fully observable vs partially obs.
  - Deterministic vs Stochastic
  - Competative vs collaborative
  - Single-agent vs multi-agent
  - Static vs Dynamic
  - Descrete vs. Continous
  - Episodic vs. Sequential
  - Known vs unknown.

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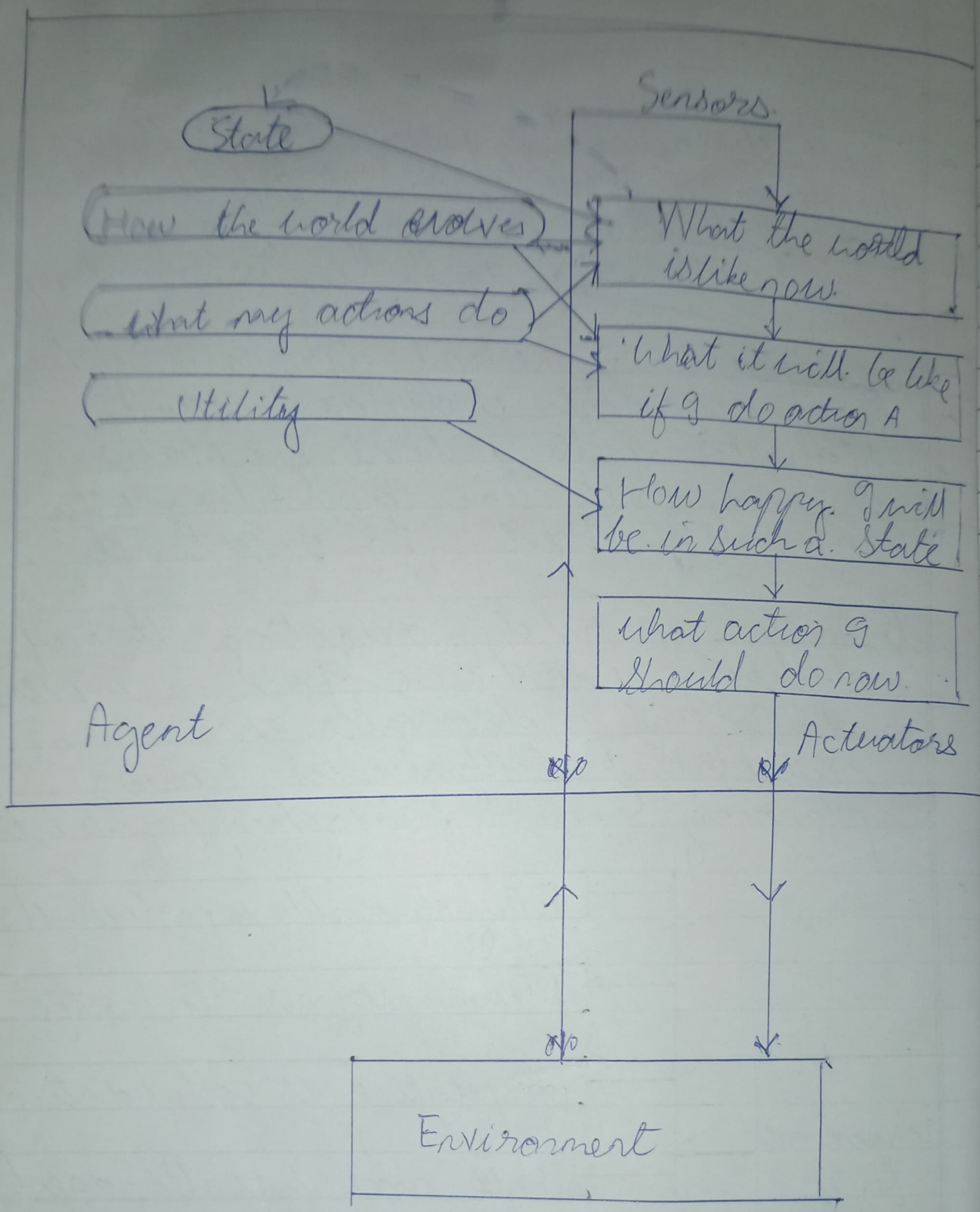
A-3)


Sensors
State
How the world evolves
What the world is like now
What my actions do
What it will be like if I do action A
Utility
How happy I will be in such a state
What action I should do now
Agent
Actuators
Environment

Structure of intelligent agents

- Intelligent agents posses four fundamental components: sensors, actuators, knowledge bases and reasoning mechanisms.

- Sensors serve to gather data from the environment, providing inputs to the agent.

- Actuators enact changes in the environment based on decisions made by the agent.

- Knowledge bases store information acquired from sensors and past experiences.

- Reasoning mechanisms process data from sensors and knowledge base to make informed decisions and take appropriate actions.

- Examples: In a self-driving car, sensors such as cameras and LiDAR collect data about the vehicles surroundings. This data is processed by the cars onboard computer, which uses its knowledge base to recognize objects like pedestrians and traffic signs. Based on this information, the reasoning mechanism determines the optimal actions for the car to take, such as accelerating

braking, or changing lanes, which are then executed by actuators.

- Overall, the interaction between these components enables intelligent agents to percieve, reason about and act upon their environments, exhibiting intelligent behaviour in various real-world scenarios.

A-4)

i.) Process of problem-solving to achieve specific goals.

-) Problem solving agents identify problems, formulate them and search for solutions.

-) They analyze problems spaces, apply search algorithms and evaluate potential solutions.

ii). Role of problem-solving agents:

-). Problem-solving agents aim to find solutions to achieve predefined goals.

-). They analyze problem instances and apply problem-solving strategies

-) Examples: route planning agents, puzzle solving agents, and theorem proving agents

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iii) Formulation of problems :

- Problems are defined by initial States, actions, transition models, and goal States.
- Problem - solving agents represents problems in a suitable formalism.
- Example: In the 8 puzzle problem, the initial State is a scrambled puzzle configuration actions are moving tiles, and the goal state in a specific arrangement.

iv) Problem - solving approach :

- Agents analyze problem spaces through search algorithms like BFS, DFS
- They explore states, apply actions and evaluate resulting states.
- Example: In the "Travelling Salesman Problem", agents explore possible routes to minimize travel distance.

V) Stratagies employed by agents;

•) Heuristic search uses domains-specific knowledge to guide search.

•) Adversarial search considers opponents moves in competitive settings.

Example: chess playing agents use heuristic search to evaluate potential moves based on board positions